

Practical – 1

Practical 1 AIM:

Linux System Administration and File Management for Marketing Analytics Team

Problem Definition :

Scenario: You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers Where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team.

Tasks:

1. Identify the current user.
2. Find the current working directory.
3. Display system information.
4. Check system uptime and date.
5. Create a directory named MarketingAnalytics.
6. Create subdirectories: RawData, ProcessedData, Reports.
7. Create data and report files.
8. Add sample content.
9. Display file contents.
10. List all files and directories.
11. Search for report files.
12. Search specific content inside files.
13. Copy report files to Backup directory.
14. Rename files.
15. Move files between directories.
16. Delete a file.
17. Delete an empty directory.

Output:

```
bhumi@kali: ~/MarketingAnalytics
Session Actions Edit View Help

(bhumi@kali)-[~]
$ whoami
bhumi

(bhumi@kali)-[~]
$ pwd
/home/bhumi

(bhumi@kali)-[~]
$ uname -a
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05)
) x86_64 GNU/Linux

(bhumi@kali)-[~]
$ uptime
date
23:23:03 up 20 min, 1 user, load average: 0.13, 0.31, 0.29
Thursday 10 September 2026 11:23:03 PM IST

(bhumi@kali)-[~]
$ mkdir MarketingAnalytics
cd MarketingAnalytics

(bhumi@kali)-[~/MarketingAnalytics]
$ mkdir RawData ProcessedData Reports

(bhumi@kali)-[~/MarketingAnalytics]
$ touch RawData/customer_data.txt Reports/sales_report.txt

(bhumi@kali)-[~/MarketingAnalytics]
$ cat << EOF > RawData/customer_data.txt
101. Raj, Ahmedabad
102. Priya, Vadodara
103. Amit, Surat
EOF

(bhumi@kali)-[~/MarketingAnalytics]
$ echo "Monthly Sales Report - Marketing Analytics Team" > Reports/sales_report.txt

(bhumi@kali)-[~/MarketingAnalytics]
$ cat RawData/customer_data.txt
cat Reports/sales_report.txt
101, Raj, Ahmedabad
102, Priya, Vadodara
103, Amit, Surat
Monthly Sales Report - Marketing Analytics Team

(bhumi@kali)-[~/MarketingAnalytics]
$ ls -la
ls -R
total 20
drwxrwxr-x 5 bhumi bhumi 4096 Sep 10 23:23 .
drwx----- 23 bhumi bhumi 4096 Sep 10 23:23 ..
drwxrwxr-x 2 bhumi bhumi 4096 Sep 10 23:23 ProcessedData
drwxrwxr-x 2 bhumi bhumi 4096 Sep 10 23:23 RawData
drwxrwxr-x 2 bhumi bhumi 4096 Sep 10 23:23 Reports
.:
ProcessedData RawData Reports

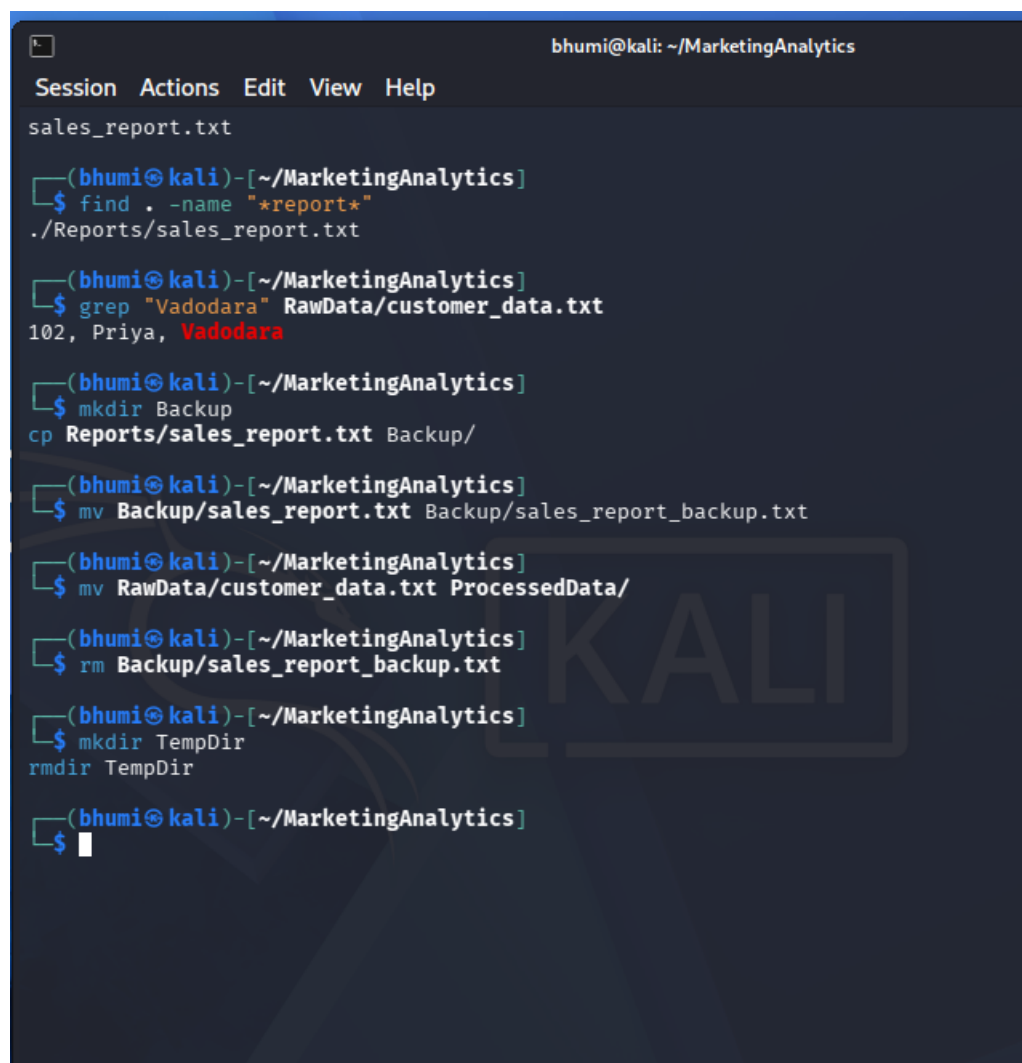
./ProcessedData:

./RawData:
customer_data.txt

./Reports:
sales_report.txt

(bhumi@kali)-[~/MarketingAnalytics]
$ find . -name "*report*"
./Reports/sales_report.txt

(bhumi@kali)-[~/MarketingAnalytics]
```



```
bhumi@kali: ~/MarketingAnalytics
Session Actions Edit View Help
sales_report.txt

(bhumi@kali)-[~/MarketingAnalytics]
$ find . -name "*report*"
./Reports/sales_report.txt

(bhumi@kali)-[~/MarketingAnalytics]
$ grep "Vadodara" RawData/customer_data.txt
102, Priya, Vadodara

(bhumi@kali)-[~/MarketingAnalytics]
$ mkdir Backup
cp Reports/sales_report.txt Backup/

(bhumi@kali)-[~/MarketingAnalytics]
$ mv Backup/sales_report.txt Backup/sales_report_backup.txt

(bhumi@kali)-[~/MarketingAnalytics]
$ mv RawData/customer_data.txt ProcessedData/

(bhumi@kali)-[~/MarketingAnalytics]
$ rm Backup/sales_report_backup.txt

(bhumi@kali)-[~/MarketingAnalytics]
$ mkdir TempDir
rmdir TempDir

(bhumi@kali)-[~/MarketingAnalytics]
$
```

Key Questions / Analysis

Q1. What is the difference between absolute and relative paths?

Answer: Absolute path: Starts from the root directory (/) and gives the complete location of a file or folder. Relative path: Starts from the current working directory and gives the location relative to it.

Q2. Why are Linux permissions important for file security?

Answer: Linux permissions control who can read, write, or execute files, preventing unauthorized access and protecting data.

Q3. What is the purpose of the grep command?

Answer: The grep command is used to search for specific text or patterns within files and display the matching lines.

Q4. How does Linux support efficient file management?

Answer: Linux provides commands to create, copy, move, rename, search, and organize files and directories efficiently.

Q5. Why are backups important in enterprise environments?

Answer: Backups protect important data from accidental deletion, hardware failures, cyberattacks, and disasters, ensuring quick data recovery and business continuity.

Supplementary Problems

1. Create a shell script to automate project directory creation.
2. Find the top 10 largest files in the system.
3. Schedule a backup task using cron.
4. Monitor CPU utilization continuously using top/htop.
5. Generate a disk usage report for all project directories.

Output:

```

user123@Ubuntu2:~/Desktop/MarketingAnalysis$ cd ..
user123@Ubuntu2:~/Desktop$ mkdir -p Project/{RawData,ProcessData,Reports,Backup,Scripts,Logs}
user123@Ubuntu2:~/Desktop$ cd Project
user123@Ubuntu2:~/Desktop/Project$ ls -R Project-
ls: cannot access 'Project-': No such file or directory
user123@Ubuntu2:~/Desktop/Project$ ls -R Project
ls: cannot access 'Project': No such file or directory
user123@Ubuntu2:~/Desktop/Project$ cd ..
user123@Ubuntu2:~/Desktop$ ls -R Project
Project:
Backup Logs ProcessData RawData Reports Scripts

Project/Backup:

Project/Logs:

Project/ProcessData:

Project/RawData:

Project/Reports:

Project/Scripts:
user123@Ubuntu2:~/Desktop$ find / -type f -exec du -h {} + 2>/dev/null | sort -hr | head -10
607M    /var/lib/snapd/snaps/gnome-46-2404_153.snap
395M    /var/lib/snapd/snaps/mesa-2404_1165.snap
274M    /var/lib/snapd/snaps/firefox_8107.snap
169M    /snap/firefox/8107/usr/lib/firefox/libxul.so
155M    /snap/mesa-2404/1165/usr/lib/i386-linux-gnu/libLLVM.so.20.1
140M    /snap/mesa-2404/1165/usr/lib/i386-linux-gnu/libLLVM.so.19.1
137M    /snap/mesa-2404/1165/usr/lib/x86_64-linux-gnu/libLLVM.so.20.1
133M    /snap/mesa-2404/1165/usr/lib/i386-linux-gnu/libLLVM-17.so.1
130M    /usr/lib/x86_64-linux-gnu/libLLVM.so.21.1
123M    /snap/mesa-2404/1165/usr/lib/x86_64-linux-gnu/libLLVM.so.19.1

```

```

user123@Ubuntu2:~/Desktop$ crontab -e
no crontab for user123 - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 0 2 * * * cp -r ~/Project ~/Project_Backup
Choose 1-3 [1]: 1
crontab: installing new crontab
user123@Ubuntu2:~/Desktop$ 0 2 * * * cp -r ~/Project ~/Project_Backup
0: command not found
user123@Ubuntu2:~/Desktop$ crontab -l
user123@Ubuntu2:~/Desktop$ top

```

```

top - 16:39:08 up 57 min, 1 user, load average: 1.57, 0.75, 0.57
Tasks: 279 total, 1 running, 278 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.3 sy, 0.0 ni, 99.6 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 8269.4 total, 1055.3 free, 3296.3 used, 3810.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 4973.1 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3354	user123	20	0	6249328	468960	284632	S	4.3	5.5	7:14.26	gnome-shell
7740	user123	20	0	3579000	415172	146632	S	0.6	4.9	1:21.37	ptxixis
4578	user123	20	0	11.6g	675932	261592	S	0.3	8.0	4:56.16	firefox
4950	user123	20	0	2674232	173520	104432	S	0.3	2.0	0:15.40	Privileged Cont
5859	user123	20	0	2582236	75632	60064	S	0.3	0.9	0:05.05	Web Content
7242	root	20	0	0	0	0	I	0.3	0.0	0:01.79	kworker/u27:5-events_unbound
8597	user123	20	0	11.6g	722600	269704	S	0.3	8.5	3:01.34	firefox
1	root	20	0	25908	17096	11612	S	0.0	0.2	0:09.16	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u24:0-ipv6_addrconf
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.20	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:03.45	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.51	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.26	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kprobe-optimizer
20	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
23	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
24	root	rt	0	0	0	0	S	0.0	0.0	0:00.14	migration/1
25	root	20	0	0	0	0	S	0.0	0.0	0:19.99	ksoftirqd/1
27	root	0	-20	0	0	0	I	0.0	0.0	0:00.49	kworker/1:0H-kblockd
28	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/2

```

0[| 1.9%] 3[| 0.6%]
1[| 0.3%] 4[| 1.3%]
2[| 0.0%] 5[| 0.6%]
Mem[|||||||||||||||||||||||||||||||||||||]3.42G/8.08G Tasks: 153, 1033 thr, 135 kthr; 6 running
Swp[|||||] Load average: 1.48 0.87 0.63
Uptime: 00:59:13

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
3354 user123 20 0 6165M 458M 199M S 2.3 5.5 3:34.97 /usr/bin/gnome-shell --mode=ubuntu
12401 user123 20 0 21588 6724 3960 R 1.3 0.1 0:00.30 htop
7764 user123 20 0 3496M 406M 0 S 0.6 4.9 0:08.46 /usr/bin/ptxixis --gaplication-service
3387 user123 20 0 6165M 458M 0 S 0.3 5.5 0:22.38 /usr/bin/gnome-shell --mode=ubuntu
3389 user123 20 0 6165M 458M 0 S 0.3 5.5 0:28.82 /usr/bin/gnome-shell --mode=ubuntu
3391 user123 20 0 6165M 458M 0 S 0.3 5.5 0:29.26 /usr/bin/gnome-shell --mode=ubuntu
3392 user123 20 0 6165M 458M 0 S 0.3 5.5 0:30.64 /usr/bin/gnome-shell --mode=ubuntu
3393 user123 20 0 6165M 458M 0 S 0.3 5.5 0:31.77 /usr/bin/gnome-shell --mode=ubuntu
3394 user123 20 0 6165M 458M 0 S 0.3 5.5 0:34.35 /usr/bin/gnome-shell --mode=ubuntu
3405 user123 20 0 6165M 458M 0 S 0.3 5.5 0:16.10 /usr/bin/gnome-shell --mode=ubuntu
4578 user123 20 0 11.6G 668M 255M S 0.3 8.0 1:27.92 /snap/firefox/8107/usr/lib/firefox/firefox
4793 user123 20 0 11.6G 668M 0 S 0.3 8.0 0:11.31 /snap/firefox/8107/usr/lib/firefox/firefox
5129 user123 20 0 2579M 135M 97316 S 0.3 1.6 0:04.60 /snap/firefox/8107/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle
5135 user123 20 0 2579M 135M 0 S 0.3 1.6 0:00.66 /snap/firefox/8107/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle
5741 user123 20 0 3041M 497M 0 S 0.3 6.0 0:00.18 /snap/firefox/8107/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle
5876 user123 20 0 2521M 75632 0 S 0.3 0.9 0:01.60 /snap/firefox/8107/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle
7740 user123 20 0 3496M 406M 144M S 0.3 4.9 0:25.67 /usr/bin/ptxixis --gaplication-service
7760 user123 20 0 3496M 406M 0 S 0.3 4.9 0:07.25 /usr/bin/ptxixis --gaplication-service
7762 user123 20 0 3496M 406M 0 S 0.3 4.9 0:07.42 /usr/bin/ptxixis --gaplication-service
7763 user123 20 0 3496M 406M 0 S 0.3 4.9 0:07.82 /usr/bin/ptxixis --gaplication-service
8644 user123 20 0 11.6G 702M 0 S 0.3 8.5 0:06.83 /snap/firefox/8107/usr/lib/firefox/firefox --profile /home/user123/snap/firefox/co
9476 user123 20 0 2987M 396M 114M S 0.3 4.8 1:08.75 /snap/firefox/8107/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle
12298 user123 20 0 6165M 458M 0 S 0.3 5.5 0:00.01 /usr/bin/gnome-shell --mode=ubuntu
1 root 20 0 25908 17096 11612 S 0.0 0.2 0:09.18 /usr/lib/systemd/systemd --switched-root --system --deserialize=51
659 root 19 -1 50928 20048 18200 S 0.0 0.2 0:01.30 /usr/lib/systemd/systemd-journald
701 systemd-oo 20 0 10808 7808 6740 S 0.0 0.1 0:00.93 /usr/lib/systemd/systemd-oomd
702 systemd-re 20 0 22804 15332 12412 S 0.0 0.2 0:01.73 /usr/lib/systemd/systemd-resolved

?help F2Setup F3Search F4Filter F5Free F6SortBy F7Nice F8Nice F9Kill F10Quit

user123@Ubuntu2:~/Desktop/OS$ cd ..
user123@Ubuntu2:~/Desktop$ du -sh */
24K MarketingAnalysis/
192K OS/
20K Project/
user123@Ubuntu2:~/Desktop$

```

Key Skills to be Addressed

- Linux Command Line Operations
- File and Directory Management
- Text Processing and Searching
- User and Permission Management
- Process Monitoring

Tools /Technology

Ubuntu Linux / CentOS Linux

Linux Terminal

Bash Shell